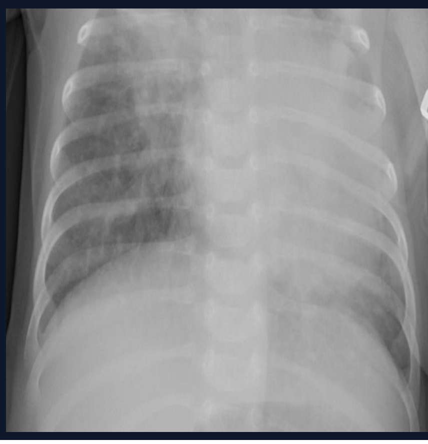
RADINOVA AI CLINICAL SUITE

Multi-Modal Neural Diagnostic Decision Support

STUDY TYPE: Chest Radiography (X-Ray)
DATE / TIME: 2026-08-23 01:48 UTC
STATUS: ANALYSIS COMPLETED

Patient Name: Eleanor Vance	**Patient ID:** RN-2026-00142	**Attending:** Dr. Alexander Ross, MD	**Facility:** RadiNova Medical Center



PRIMARY AI MODEL OUTPUT:
PNEUMONIA CONSOLIDATION DETECTED

DIAGNOSTIC CONFIDENCE:
100.0%

Inference Model: PyTorch DenseNet-121 (Chest Radiography)
Explainability Layer: Grad-CAM (denseblock4.denselayer16)

Modality Gatekeeper: PASSED (100.0% Anat. Confidence)
Class Distribution: NORMAL: 0.0% • **PNEUMONIA:** 100.0%

ANATOMICAL ASSESSMENT & RADIOLOGIC SIGNS

Anatomical Zone Breakdown	Pathological Signs Evaluated
• **Right Upper Lobe:** Trace Infiltrate (Involvement: 18%) • **Right Mid Zone (Perihilar):** Active Infiltrate (Involvement: 45%) • **Right Lower / Costophrenic:** Focal Density (Involvement: 65%) • **Left Lung Field:** Clear Aeration (Involvement: 0%)	• **Air Bronchograms:** [PRESENT] — Tubular radiolucency within alveolar opacification • **Silhouette Sign:** [PRESENT] — Loss of normal radiographic heart or diaphragm borders • **Kerley B Lines:** [ABSENT] — Short horizontal interstitial markings at pleural margin • **Costophrenic Blunting:** [PRESENT] — Trace reactive parapneumonic fluid collection

CLINICAL DECISION SUPPORT & DIFFERENTIAL CONSIDERATIONS

Clinical Summary: Marked radiographic consolidation and pulmonary opacity consistent with pneumonia pattern.

Differential Considerations:

- Bacterial lobar pneumonia (e.g., Streptococcus pneumoniae)
- Atypical / Viral bronchopneumonia
- Pulmonary edema / alveolar infiltrate

Recommended Follow-Up & Clinical Correlation:

- Correlate with clinical presentation (fever, purulent sputum, auscultatory crackles)
- Evaluate inflammatory markers (Complete Blood Count, CRP, procalcitonin)
- Consider microbiological sputum cultures and blood cultures prior to targeted antimicrobial therapy
- Follow-up posteroanterior (PA) chest radiograph in 4-6 weeks to assess resolution

Attending Radiologist / Clinician Verification	**Clinical Governance Notice**
Physician: **Dr. Alexander Ross, MD** Signature: ____________________________ Date: ________________________________	AI-assisted prediction / decision support — requires review by a qualified healthcare professional. Model outputs and Grad-CAM explainability maps serve as secondary clinical decision support (CDSS) and do not substitute for certified clinical judgment.